Is Bitcoin Chaotic?

Nonlinear and Chaos Analysis of BTC Returns and Volatility

Phase-space reconstruction, chaos tests and surrogate data

Anuj Kothari

November 3, 2026

Outline

1. Question and Method
2. Part A: BTC Returns
3. Part B: Returns vs Volatility
4. Conclusions

1. The Question

Two questions, kept separate

1. **Nonlinearity:** is there dependence that a linear Gaussian process cannot produce?
2. **Deterministic chaos:** is that nonlinearity generated by a low-dimensional deterministic system, rather than by randomness?

A series can be clearly **nonlinear** and still be **random**: volatility clustering (GARCH) is the standard example.

Why care

Chaos is deterministic, so it is **predictable over short horizons**. Randomness is not. Finding chaos in BTC would mean nonlinear forecasting models should work.

Ground rules

No plot and no single test may decide "chaos". Every statistic is compared with **surrogate** series that are random by construction.

2. Data and the Four Series

BTC daily close (Binance), 2017-08-17 to 2026-09-02, $n \approx 3300$ days. The data file is only read.

Series	Definition	Measures
returns	$r_t = \ln P_t - \ln P_{t-1}$	direction and size of the move
absolute returns	$\lvert r_t \rvert$	daily volatility
squared returns	r_t^2	daily variance
20-day realised volatility	$RV_t = \sqrt{\frac{1}{20}\sum_{i=0}^{19} r_{t-i}^2}$	smoothed volatility

- Part A: returns only (notebook `chaos_classification`).
- Part B: all four series side by side (notebook `volatility_chaos`).

3. Toolkit in Plain Words

Tool	What it asks
Return map, phase space	Do the points form a thin, folded shape (attractor) or a blob (noise)?
Recurrence plot	When the series revisits a state, does it then follow the same path (long diagonal lines)?
Average Mutual Information	After how many days (τ) does today stop telling us much about the future?
False Nearest Neighbours	How many dimensions (m) are needed to unfold the dynamics? Noise never unfolds.
Largest Lyapunov exponent	Do nearly identical states drift apart exponentially?
Correlation dimension D_2	Is the data on a low-dimensional object (D_2 stops growing with m)?
0–1 test	Regular ($K \approx 0$) or irregular ($K \approx 1$)? Noise also gives $K \approx 1$.
Time-reversal asymmetry	Does the series look different when played backwards?
S-map gain, prediction skill	Does a local nonlinear forecast beat a linear one?

4. Surrogate Data and the Decision Rule

Surrogate	Keeps	Rejecting it means
IAAFT	distribution, spectrum	nonlinearity
GARCH(1,1)-t	fat tails, vol clustering	beyond a stochastic volatility model

- GARCH surrogates are simulated returns put through the **same** transformation as the data.
- 99 (Part A) and 999 (Part B) surrogates per family; Benjamini–Hochberg FDR over the 48 tests of Part B.

Classification rule (fixed in advance)

Periodic / quasi-periodic: $K < 0.2$ and D_2 saturates.

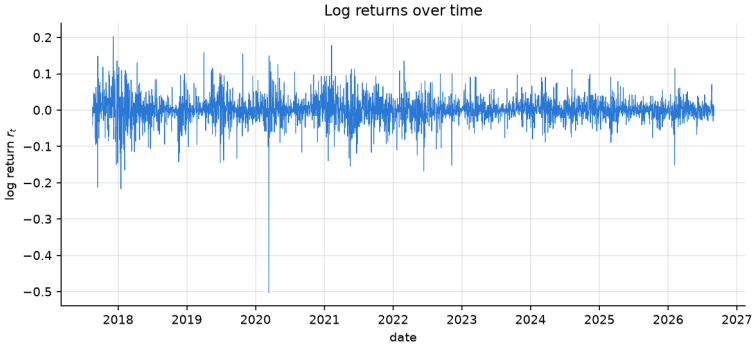
Potentially chaotic: $K > 0.8$, D_2 saturates, and ≥ 2 statistics differ from **both** IAAFT and GARCH.

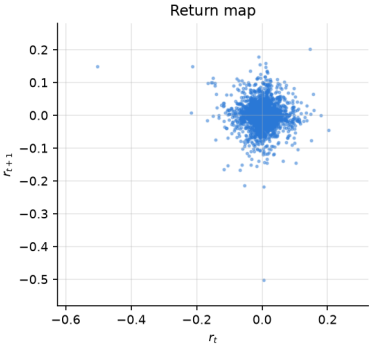
Random/Stochastic: $K > 0.8$ and nothing differs from GARCH.

Inconclusive: anything else.

A positive Lyapunov exponent alone never counts: random series give one too.

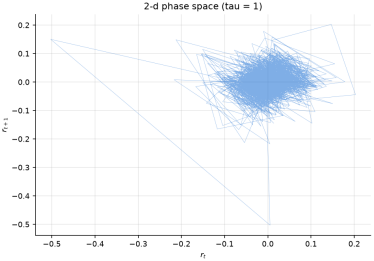
5. Returns and the Return Map

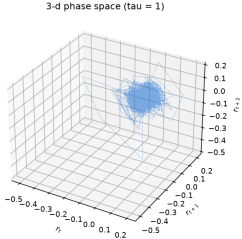


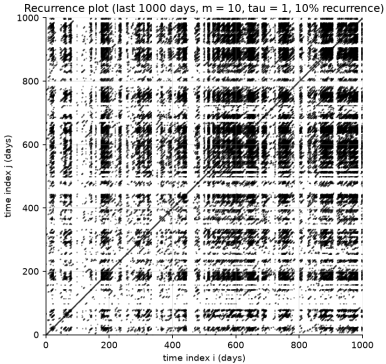


- Bursts of large moves (2020 crash, 2021 to 2022) but no visible cycle.
- Return map r_t vs r_{t+1} is a round cloud: no thin curve of a deterministic map.

6. Phase Space and Recurrence

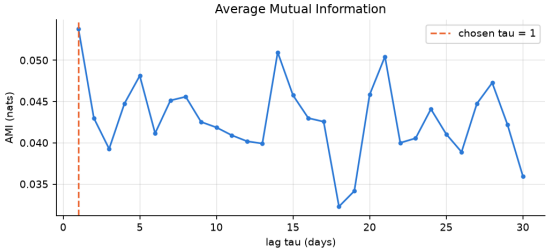




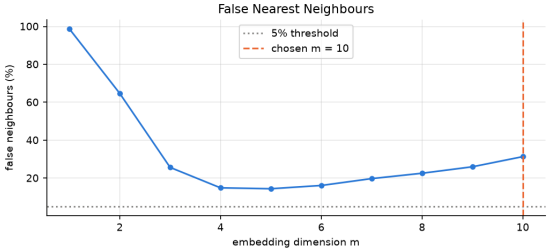


- 2-d and 3-d reconstructions ($\tau = 1$): a dense ball, no folded structure.
- Recurrence plot: square **blocks** (calm and turbulent periods), not long diagonal lines.

7. Delay and Dimension



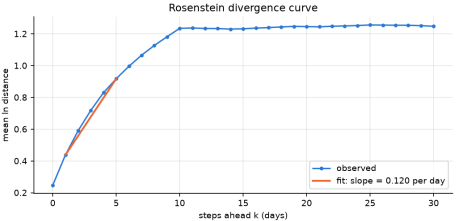
AMI is low at all lags; $\tau = 1$.

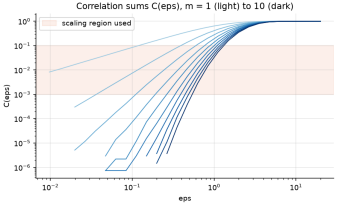


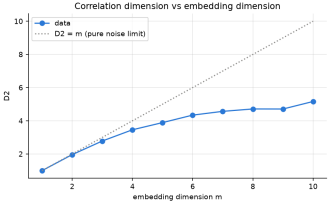
False neighbours bottom out at 14% ($m = 5$) and rise again; the 5% threshold is never reached.

The dynamics never "unfold" in up to 10 dimensions: the signature of a stochastic series.

8. Lyapunov Exponent and Correlation Dimension

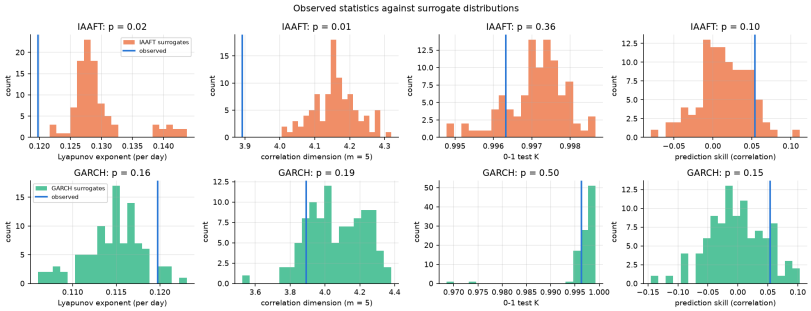






- $\lambda_1 = 0.120$ per day, $R^2 = 0.99$: positive, but surrogates give 0.11 to 0.14.
- D_2 levels off near 4.5: far too high for a simple attractor; noise of this length levels off at the same height.
- 0–1 test: $K = 0.997$ (irregular, chaos or noise).

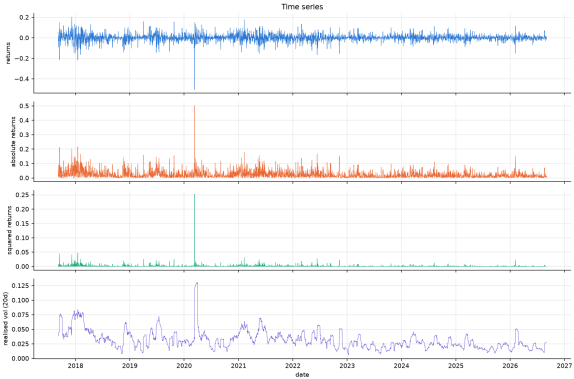
9. Surrogate Test and Classification of Returns

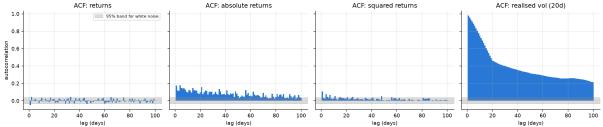


	Lyapunov	D_2 $(m = 5)$	0–1 K	prediction skill
p vs IAAFT	0.02	0.01	0.36	0.10
p vs GARCH	0.16	0.19	0.50	0.15

Returns differ from IAAFT (nonlinear) but not from GARCH. Classification: **Random/Stochastic**.

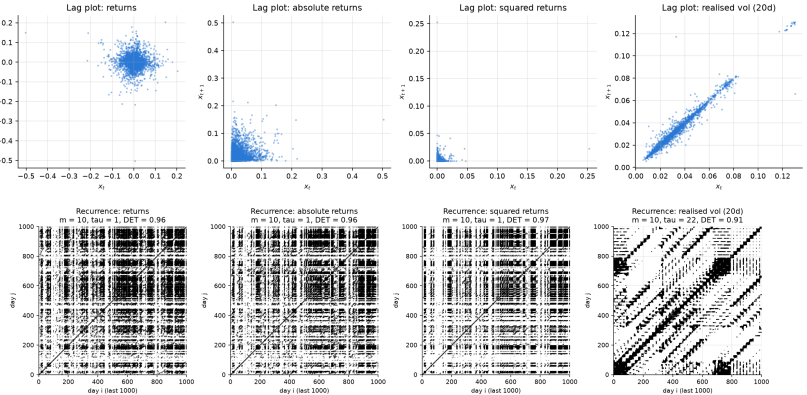
10. The Four Series and Their Memory





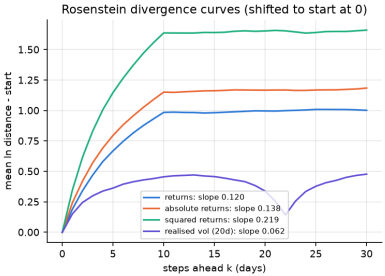
- Returns: autocorrelation close to zero (largest -0.05 at lag 1).
- Absolute and squared returns: small but very **persistent** autocorrelation (volatility clustering).
- Realised volatility: strong autocorrelation, partly built in by the overlapping 20-day window.
- KPSS rejects stationarity for all three volatility series.

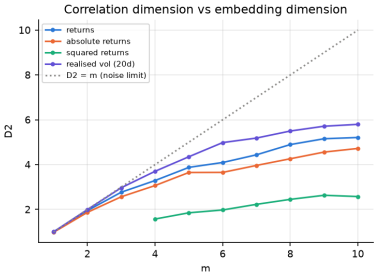
11. Lag Plots and Recurrence Plots

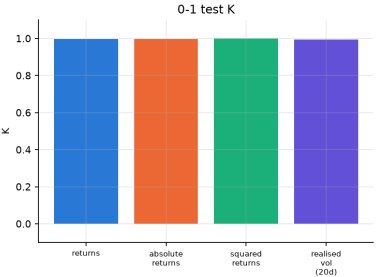


- Volatility lag plots are tilted wedges (persistence), not thin curves.
- Recurrence: blocks for daily series; diagonal bands for realised volatility caused by the 20-day smoothing.

12. Chaos Diagnostics for All Four Series

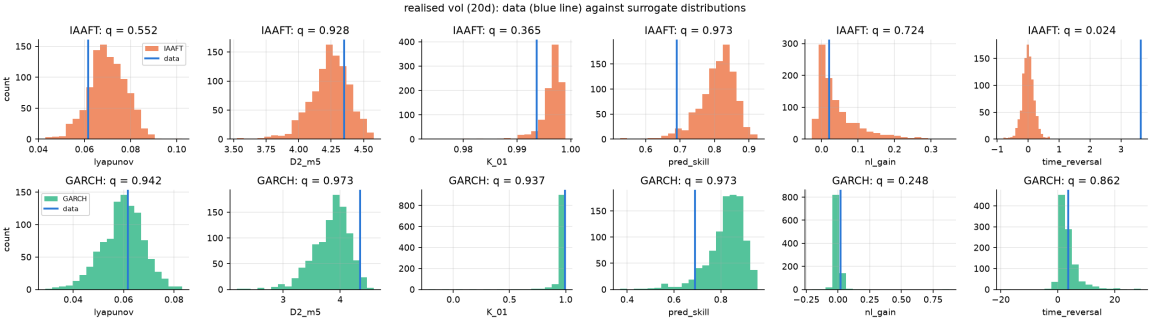






	returns	$\lvert r\rvert$	r^2	RV (20d)
τ / m (FNN 5% reached?)	1 / 10 (no)	1 / 10 (no)	1 / 10 (no)	22 / 10 (no)
Lyapunov (per day)	0.120	0.138	0.219	0.062
D_2 (levels off at)	4.1	3.8	2.6	5.2
0–1 test K	1.00	1.00	1.00	0.99

13. Realised Volatility Against Its Surrogates



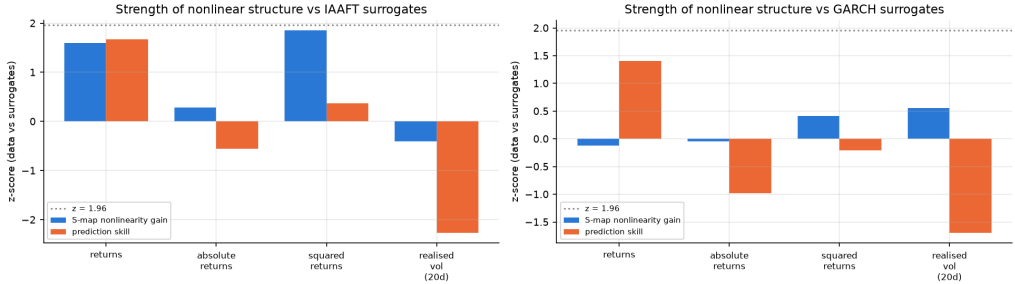
- Top (IAAFT): time-reversal asymmetry is far outside the surrogates ($z = 18.6$, $q = 0.024$).
- Bottom (GARCH): the same asymmetry is right in the middle ($z = 0.01$). Every other statistic matches too.

14. Comparison Table

	returns	$\lvert r \rvert$	r^2	RV (20d)
Differs from IAAFT (FDR)	Lyapunov, D_2, asymmetry	Lyapunov, asymmetry	S-map gain	asymmetry
Differs from GARCH (FDR)	none	none	none	none
Nonlinear	**yes**	**yes**	**yes**	**yes**
Beyond GARCH	**no**	**no**	**no**	**no**
Time-reversal z vs IAAFT	5.4	2.9	0.9	18.6
S-map gain (z vs IAAFT)	0.005 (1.6)	0.006 (0.3)	0.146 (1.9)	0.021 (-0.4)
Classification	**Random/Stoch.**	**Random/Stoch.**	**Random/Stoch.**	**Random/Stoch.**

All four series are nonlinear, none goes beyond GARCH, none meets the rule for potential chaos.

15. Is Volatility More Nonlinear Than Returns?



- **Asymmetry:** yes. Realised volatility rises fast and decays slowly ($z = 18.6$ vs 5.4 for returns).
- **Predictability:** no. Nonlinear forecasts of volatility are not better than for surrogates; against GARCH all z-scores are small.

16. Non-Stationarity: Split-Sample Check

Full test repeated on each half (GARCH refitted, 499 surrogates, FDR within each half).

Differs from IAAFT	returns	$\lvert r \rvert$	r^2	RV (20d)
Full sample	Lyap., D_2, asym.	Lyap., asym.	S-map gain	asymmetry
2017-09 to 2022-02	D_2	asymmetry	D_2	asymmetry
2022-02 to 2026-09	none	none	none	none
Differs from GARCH (any period)	none	none	none	none

The nonlinearity comes from the turbulent 2017 to 2022 regime and disappears after 2022: a sign of volatility regimes, not of a stable deterministic attractor.

17. Research Interpretation

1. **Volatility contains nonlinear dependence.** So do returns. The strongest signature is the time asymmetry of realised volatility.
2. **GARCH explains it completely.** No statistic of any series differs from GARCH(1,1)-t surrogates (smallest $q = 0.22$).
3. **No convincing evidence of deterministic chaos.** FNN never converges, D_2 levels off only at 2.6 to 5.2 like noise, $\lambda_1 > 0$ and $K \approx 1$ also hold for the surrogates.
4. **Classification:** returns, absolute returns, squared returns and realised volatility are all **Random/Stochastic**: nonlinear, but stochastic.
5. **Time variation:** the nonlinear structure is concentrated in 2017 to 2022.

BTC behaves like a stochastic volatility process. Nonlinearity is real; chaos is not supported.

18. Limits and Next Steps

Limits

- About 3300 daily points: enough for low-dimensional chaos, not for high-dimensional chaos.
- Chaos tests lose power quickly with noise.
- Only one asset and one frequency (daily).

Next steps

1. Test realised volatility against richer stochastic nulls: asymmetric GARCH (GJR, EGARCH), long memory (HAR, FIGARCH), regime switching.
2. Repeat with intraday realised volatility.
3. Extend to other cryptocurrencies and Nifty 50 stocks.

Thank you

Questions and suggestions welcome.